

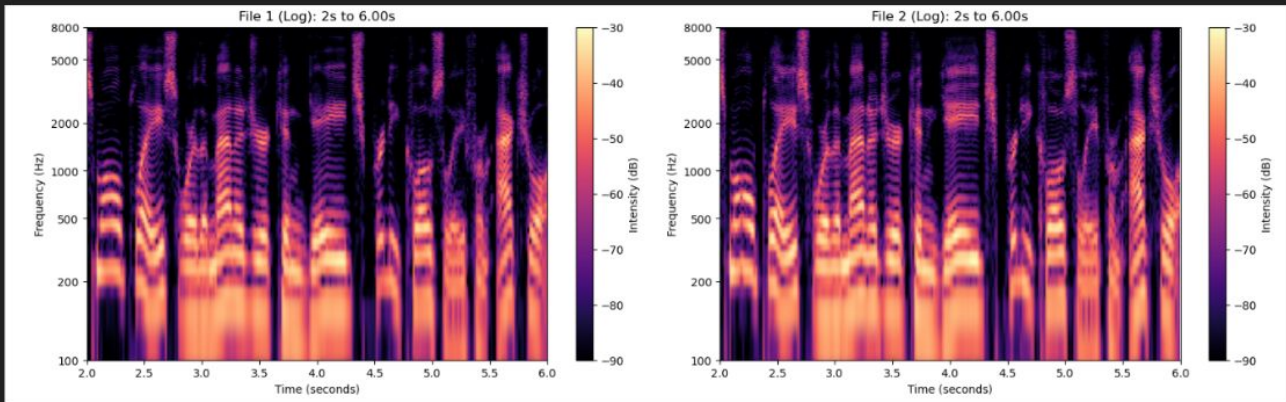
Detective Spoof

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Background and Motivation

Problem: AI Voice cloning can generate realistic sounding speech

- > Impersonation scams phishing calls
- > identity fraud
- > fake emergency calls
- Goal: Detect whether an input audio clip is real or fake
- Speech/audio waveform data



Real

Fake

Existing Methods

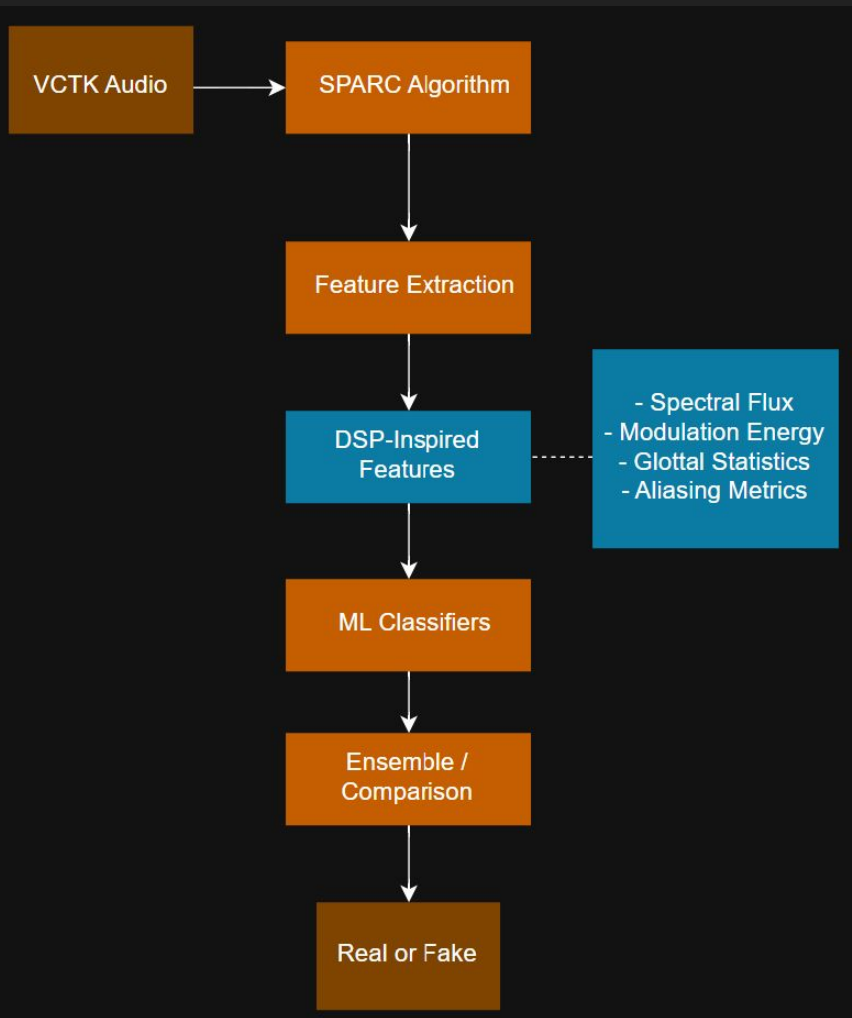


- Spectral Analysis - Detect unnatural speech patterns
 - Helps humans spot anomalies but labor intensive
- MFCC-Based Methods - Extract compact speech features
 - Common in speech recognition
- Deep Learning Models - models learn patterns automatically
 - Often difficult to interpret
- Commercial Systems (Resemble AI) - Detect breathing irregularities and synthesis artifacts

→ Most methods rely on spectral and temporal inconsistencies in synthetic Speech

Key Differences

- Interpretable
DSP-inspired features
- Spoof artifact analysis
- Ensemble evaluation



Voice Conversion

We used Speech Articulatory Coding (SPARC) for getting converted audio.

Available for offline conversion, produces very convincing results at 16khz.

Developed by Berkeley Speech Group

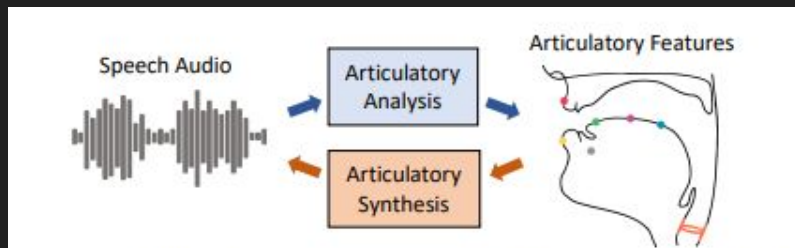


Fig. 1. The Speech Articulatory Coding (SPARC) framework. It encodes speech using articulatory features (analysis) and decodes these features back into speech (synthesis).

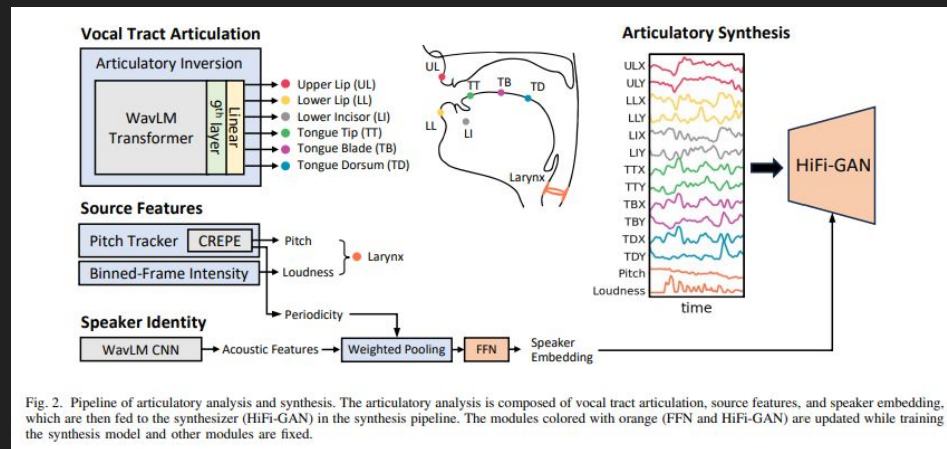


Fig. 2. Pipeline of articulatory analysis and synthesis. The articulatory analysis is composed of vocal tract articulation, source features, and speaker embedding, which are then fed to the synthesizer (HiFi-GAN) in the synthesis pipeline. The modules colored with orange (FFN and HiFi-GAN) are updated while training the synthesis model and other modules are fixed.

Github: <https://github.com/Berkeley-Speech-Group/Speech-Articulatory-Coding>

Paper: <https://arxiv.org/pdf/2406.12998>

Our Method

1. Download 2k audio clips from VCTK. Set A (1k source) and Set B (1k target)
2. Use SPARC to do voice conversion to get 1k fake clips (We call this Set C)
 - a. A -> B -> C
3. Extract features from real and converted audio
4. Train ML Model
5. Analyze Sets A/C
6. Download 500 clips from VCTK Set D(source) and 500 from VCTK Set E(target)
7. Use SPARC to obtain set F converted
 - a. D -> E -> F
8. Analyze Sets C/F

Metrics

Metric	Purpose
Accuracy	Overall spoof-detection performance
Precision	Fraction of predicted spoof clips that were actually spoofed
Recall	Ability to correctly detect spoofed speech
F1-Score	Balance between precision and recall

Feature Extraction / DSP Features

Feature	Purpose
Spectral Flux	Measures frame-to-frame spectral variation
Modulation Energy	Captures temporal smoothness patterns
Glottal Statistics	Analyzes speech excitation irregularities
Aliasing Metrics	Detects synthesis/vocoder artifacts

Results

VCTK 1000 Pair

Model	Accuracy
Logistic Regression	92.8%
XGBoost	99.1%
AVG Ensemble	96.5%

VCTK 500 Pair

Model	Accuracy
Logistic Regression	94.2%
XGBoost	88.5%
AVG Ensemble	95.0%

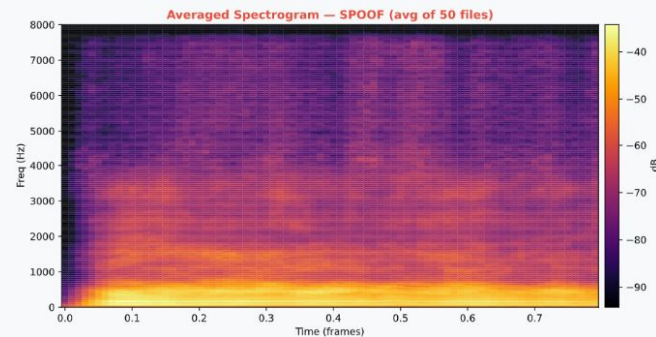
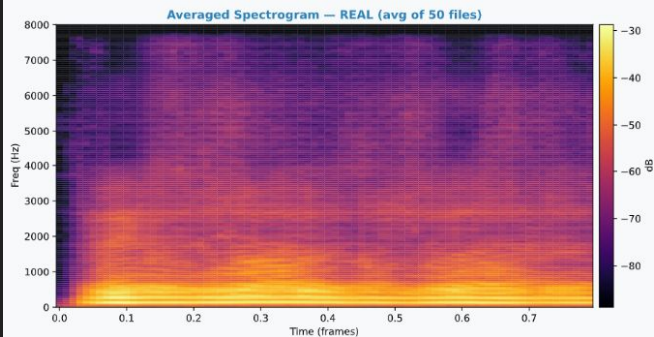
DSP Feature Analysis

Bigger number = better predictor

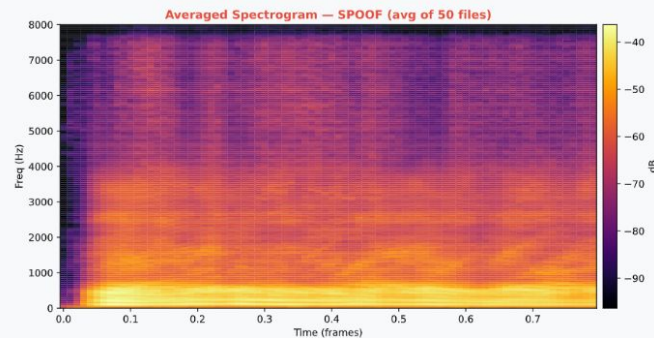
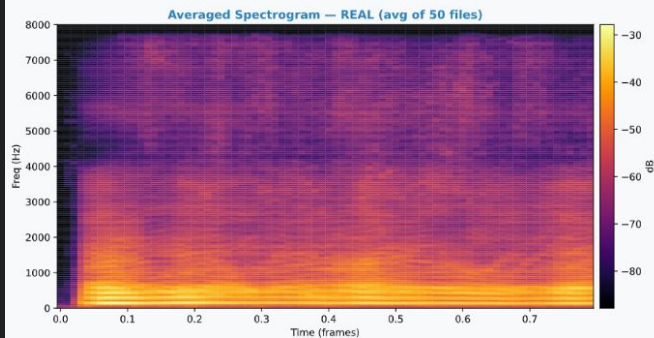


Spectrograms

Averaged Signal Analysis — VC 500 Pairs (50 files sampled per class)



Averaged Signal Analysis — VCTK 1000 Pairs (50 files sampled per class)



Comparison to Commercial Detection Approaches

[Resemble.AI](#)

Future Improvements

Key Takeaways:

- DSP-inspired features effectively detected spoofed speech
- XGBoost achieved 99.1% accuracy on the VCTK dataset
- Spectral and temporal artifacts strongly differentiated spoofed speech

Future Work

- Include additional spoof-generation methods into training set
- Test on real-world voice cloning systems
- Explore real-time spoof detection
- Combine DSP features with deep learning models